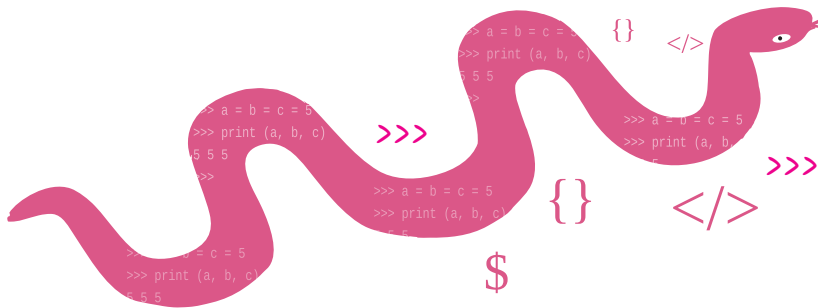


Python: A Few Useful Tips for Beginners

Python is TIOBE's programming language of the year for 2020 and is also ranked number one as per the PYPL (Popularity of Programming Language) site. The language is easy to learn, read and understand. The Python tips given in this article will help readers to write code that is concise and easy to understand.



The Python programming language, a general-purpose programming language created by Guido Van

Rossum in the late 1980s, is a free and open source language incorporating multiple programming paradigms. Rossum was motivated to create this language as he was not happy with both the C language and the UNIX shell, the tools he had access to at that time. The C language forced him to manually manage memory and the language did not have enough libraries of reusable code. UNIX shells could not handle complex logic and were slow in execution.

Python is easy to learn, read and understand, and is ideal for beginners to learn and start coding in a few hours. It has an interpreter which makes it easy to test even a single line of code quickly. The Python language is versatile, flexible, extensible and powerful. Open source libraries are available for use in various areas including Big Data, machine learning, deep learning, data analytics, simulation and scientific programming, image processing,

natural language processing, robotics, face recognition, numerical computation, data visualisation, Web development, etc.

Suitable for day-to-day tasks, Python reduces the software development time sharply by three to five times compared to other popular languages like C/C++ and Java. It supports dynamic typing and does automatic memory management. Python is TIOBE's programming language of the year 2020 and is now ranked number one as



Figure 1: Guido Van Rossum
(Link: <https://images.app.goo.gl/6BmEikkYrD2rywva9>;
Source: <https://commons.wikimedia.org>)

per the PYPL (Popularity of Programming Language) site.

I have compiled a few tips and tricks of the Python programming language which will be useful for beginners. Some of the tips are related to the

way Python allows some common tasks and constructs to be written differently. This makes the code concise and easier to read. Some other tips give a way to do some tasks using a different method instead of the usual one.

Multiple assignments

Python allows multiple assignments in a single statement. The first value on the right is assigned to the first variable on the right, and so on.

```
>>> a, b = 1, 2
>>> print(a, b)
1 2
>>>
```

A single value could be assigned to multiple variables in the following way:

```
>>> a = b = c = 5
>>> print(a, b, c)
5 5 5
>>>
```

If multiple values are assigned to a single variable, they are assigned as a tuple:

```
>>> a = 1, 2, 3
>>> print(a)
(1, 2, 3)
>>>
```